MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPES 12ABP1A, 12ABP7A, AND 12ABP19A

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Magnetic deflection electrostatic focus, aluminized screen

PIN CONNECTIONS AND DIMENSIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Ec2	Eb2	Ehk	Rg1	Eb1	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	Meg	Vdc	ft
Maximum:	6.9	0, -200	770	13,200	±180	(see note 1)	1,100	10,000
Minimum:	5.7	-------	---	7,000	----	---	-500	---
TEST CONDITIONS:	6.3	Adjust	300	10,000	----	---	Focus	---

GENERAL:

Qualification - Required

(F) Revision

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							Min	Max	
	Qualification								
1216	Base material insulating quality	All	Zone 5 (minimum)	---	---	---	---	---	---
1141	Pressure	All		---	---	---	---	---	---
5216	Cathode illumination	All		---	---	---	---	---	---
5111	Vibration (cathode-ray tubes)	All		---	---	Width	---	2.0	mm
1331	Direct interelectrode capacitance	All	Ck to all Cg1 to all	-- ---	--- ---	C C	--- ---	6.0 10 0	pF pF
5101	Face tilt	All		---	---	---	---	---	---
----	Shock	All	See note 2	---	---	---	---	---	---
	Quality conformance inspection, part 1								
5106	Bulb, screen, and faceplate quality	All		---	---	---	---	---	---
5201	Voltage breakdown	All		---	---	---	---	---	---
5201	Voltage breakdown (magnetic type)	All		---	---	---	---	---	---
5206	Gas ratio	All	See note 7	---	---	Gr	---	0.25	---
5101	Neck straightness	All	See note 3	---	---	---	---	---	---
5221	Light output	P1A	Ib = 100 µAdc	---	---	---	125	---	ftL
5223	Modulation	P1A, P7A	Ib - 100 µAdc	---	---	Δ Ec1	---	32	Vdc
		P19A	Ib2 - 2 µAdc	---	---	Δ Ec1	---	15	Vdc
5231	Spot position (magnetic deflection	All		---	---	---	---	15	mm
5236	Zero-bias anode current (magnetic deflection)	P1A, P7A		---	---	---	---	---	---
5241	Grid cutoff voltage	All		---	---	Ec1	-33	-77	Vdc
5251	Grid No. 1 leakage	All		---	---	---	---	---	---
	Quality conformance inspection, part 2								
1105	Permanence of marking	All		---	---	---	---	---	---
1101	Secureness of base, base insert and cap	All		---	---	---	---	---	---
1301	Heater current	All		---	---	If	540	660	mA
5101	Side terminal and base alinement	All	Vacant pin position No. 3	---	---	---	---	---	---
(F) 1111	Base pin solder depth	All		---	---	---	---	---	---
5101	Neck and bulb alinement (magnetic type)	All		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							Min	Max	
	Quality conformance inspection, part 2 -continued								
----	Focusing voltage at beam current condition	P1A, P7A	Ib2=100 µAdc See note 4	---	---	---	0	300	Vdc
		P19A	Ib2=2 µAdc See note 5	---	---	---	0	400	Vdc
5221	Screens	P7A, P19A		---	---	---	---	---	---
5216	Stray emission (conventional types)	All	Eb2=13,200 Vdc; Ec2 = 770 Vdc; Eb1= 1,100 Vdc	---	---	---	---	---	---
5226	Line width A (magnetic deflection)	P1A, P7A	Ib2= 100 µAdc See note 6	---	---	---	---	0.50	mm
		P19A	Ib2= 2 µAdc See note 6	---	---	---	---	0.35	mm
5251	Heater-cathode leakage	All		---	---	---	---	15	µa
5251	Electrode-currents (grid No. 2)	All	Ec1 = 0	---	---	Ic2	-15	+15	µAdc
5251	Grid No. 2 leakage	All		---	---	---	---	---	---
	Quality conformance inspection, part 2								
	Life-test provisions	All	Group C; t = 500 hours; Eb2=13,200 Vdc; Ec2= 770 Vdc; Eb1 focus	---	---	---	---	---	---
		P1A, P7A	Ib2=60 µAdc	---	---	---	---	---	---
		P19A	Ib2 = 2 µAdc	---	---	---	---	---	---
	Cathode-ray tubes, life-test end points:								
	Line width A	P1A, P7A		---	---	---	---	0.50	mm
	Line width A	P19A		---	---	---	---	0.35	mm
	Modulation	P1A, P7A		---	---	---	---	32	Vdc
	Modulation	P19A		---	---	---	---	15	Vdc
	Grid No. 1 leakage	All		---	---	---	---	---	---
	Heater-cathode leakage			---	---	---	---	---	---
	Grid No. 2 leakage			---	---	---	---	---	---
	Stray emission			---	---	---	---	---	---

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NOTES:

1. When Ec2 is greater than 330 Vdc, Rg1 shall not exceed 0.5 meg. When Ec2 is less than 330 Vdc, Rg1 shall not exceed 1.5 meg.

2. The test shall be performed in accordance with method 205, test condition A, of MIL-STD-202.

   a. The following conditions apply:

      (1) The tube shall be mounted to the elevator table by a suitable rigid assembly. Hardwood clamps shall be used at the face plate and near the neck to secure the tubes to the assembly.

      (2) The assembly shall be such as to permit changes in tube orientation. A total of 10 shocks shall be applied to the tube in the following sequencies:

Order	Number of drops	Direction
1	2	Base to face
2	2	x+
3	2	x-
4	2	y+
5	2	y-

      (3) No potentials shall be applied during shock.

   b. After shock test, cathode-ray tubes shall meet the requirements specified and shall be subjected to the following tests:

      (1) The change in position of the undeflected focused spot from its position noted before the test (see method 5231) shall not exceed 8 mm.

      (2) The change in deflection factor for electrostatic tube types shall not exceed 5 percent (see method 5248).

      (3) Starting with a tube with no loose particles the maximum number of loose particles as a result of shocking shall be no more than a total of five loose particles greater than 0.016 inch in any one cathode-ray tube. Tubes having loose particles greater than 0.016 inch shall be vibrated with the face down in a horizontal plate at 5 G's for 1 minute. If the screen shows damage from such vibration, the tube shall be rejected.

      (4) The change in grid cutoff voltage shall not exceed 10 percent (see method 5241).

      (5) There shall be no damage to the external parts of the tube.

3. The cylinder shall be 5-inches long and have a 1.503-inch maximum diameter.

4. Focusing electrode voltage shall be measured as follows: With a standard 35-to 100-line raster, modulate grid 1 by 1-microsecond pulses to produce a polka dot pattern. Adjust the beam current to peak value of 100 microamperes. Adjust the focus voltage to obtain the best circular spot at the center of the tube face. A comparator fitted with a reticle having circles approximately the size of the spot facilitates focusing. The focusing voltage obtained in this manner is the focusing electrode voltage.

5. Use an overscanned 35-to 105-line raster pattern with not less than 1/4-inch spacing between adjacent lines; then adjust for best focus at center of tube.

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NOTES: -Continued

6. Line width A shall be obtained at the tube orientation producing poorest line width at the center of the tube. The tube position which produces the poorest line width can be found by making the raster nonlinear. This can generally be accomplished by adjustment of vertical linearity control. Measurement of line width A shall be made as follows: Compress the raster until part is merged and part is expanded. Rotate the tube slowly through 180 degrees and note spacing of raster lines, for example: If portions of expanded raster become merged, line width has become poorer; if merged portion of raster becomes expanded, line width has improved.

7. This test to be performed at the conclusion of the holding period.

Custodians:

Army - EL
Navy - EC
Air Force - 80

Preparing activity:

Air Force - 80

Review activities:

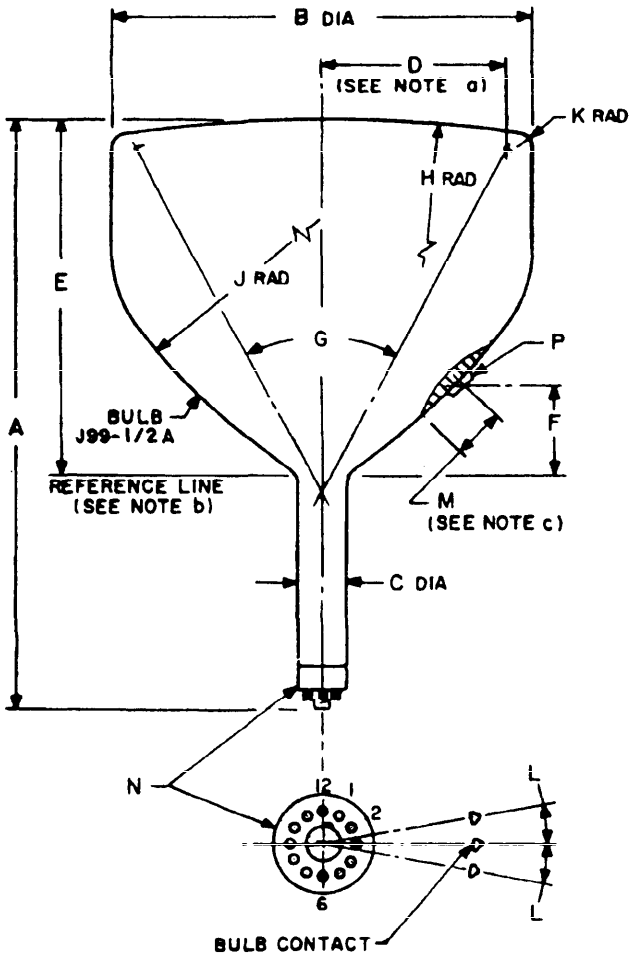
Army - EL
Navy - EC
Air Force - 11

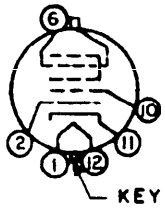
(Project 5960-2527)

User activities:

Army - MU
Navy - , MC, CG
Air Force - None

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BASING DIAGRAM

DIMENSIONS				
LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
QUALIFICATION				
L		10°		10°
N	BASE: B6-63			
P	BULB CONTACT: J1-21			
QUALITY CONFORMANCE INSPECT. PART 2				
A	17.63	18.38	447.80	466.85
B	12.31	12.56	312.67	319.02
C	1.375	1.50	34.93	38.10
D	5.50		139.70	
E	10.31	10.69	261.87	271.53
F	2.63	3.38	66.80	85.85
M	1.5		38.1	
REFERENCE DIMENSIONS (NOTE d)				
G	55°			
H	40		1016	
J	20		508	
K	.438		11.13	

PIN CONNECTIONS

Pin No.	Element
1	Heater
2	Grid No. 1
6	Focusing electrode
10	Grid No. 2
11	Cathode
12	Heater
Bulb contact	Accelerator

NOTES

a. Useful screen radius.
b. Reference line is determined by point where gage G-112, will rest on bulb cone.
c. Anticorna coating. Tube should not be handled by this part of the bulb.
d. Reference dimensions are for information only and are not required for inspection purposes.
e. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Outline drawing of electron tube types 12ABP1A, 12ABP7A and 12ABP19A

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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